Project Title	**Fitbit : Calorie Burn Prediction & Workout Pattern Clustering Using Fitbit Data**
Skills take away From This Project	**Data Preprocessing** **Feature Engineering** **Supervised Learning (Regression)** **Unsupervised Learning (Clustering)** **Dimensionality Reduction (PCA)** **Model Evaluation Metrics** **Business Interpretation of ML Results**
Domain	**Fitness Analytics / Health Tech / Machine Learning**

Problem Statement

Accurate calorie estimation during workouts is a critical component of modern fitness tracking applications such as Fitbit. While wearable sensors capture physiological signals like heart rate and session duration, additional contextual inputs such as workout type, hydration level, and user experience influence calorie expenditure.

This project focuses on building a **machine learning system** that:

1. **Predicts calories burned per workout session** using supervised regression techniques.

2. **Identifies hidden workout patterns and user segments** using unsupervised clustering techniques.

By the end of this project, learners will understand how to combine **predictive modeling** and **pattern discovery** to build intelligent, real-world fitness applications.

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Business Use Cases

- **Wearable Fitness Apps**: Real-time calorie burn prediction during workouts

- **Personalized Fitness Coaching**: Workout intensity and duration recommendations

- **Health Monitoring Platforms**: Energy expenditure insights for nutrition planning

- **User Segmentation**: Identifying workout behavior patterns without labels

- **Product Optimization**: Feature insights for fitness device manufacturers

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Dataset Description

DataSet

Dataset Columns

Column Name	Description	Source
Age	User age (years)	Manual
Gender	Male / Female	Manual
Weight (kg)	Body weight	Manual / Sensor
Height (m)	Body height	Manual
BMI	Body Mass Index	Derived
Fat_Percentage	Body fat percentage	Manual / Sensor
Max_BPM	Maximum heart rate	Sensor
Avg_BPM	Average heart rate	Sensor
Resting_BPM	Resting heart rate	Sensor
Session_Duration (hours)	Workout duration	Sensor
Workout_Type	Cardio / Strength / HIIT / Yoga	Manual
Water_Intake (liters)	Hydration level	Manual
Workout_Frequency (days/week)	Weekly workout count	Manual
Experience_Level	Beginner / Intermediate / Advanced	Manual
Calories_Burned	Calories burned per session	Target

Project Tasks Overview

Task 1: Supervised Learning – Calorie Burn Prediction (Regression)

Target Variable:

- `Calories_Burned` (continuous numeric)

ML Models to Train and Compare:

- Linear Regression/Ridge Regression/Lasso Regression
- K-Nearest Neighbors (KNN) Regressor
- XGBoost Regressor
- Decision Tree Regressor
- Random Forest Regressor
- Support Vector Regression (SVR)
- Hyperparameter tuning may be applied as an optional step to further optimize model performance and improve generalization.

Data Preprocessing:

- Handle missing values
- Outlier detection and capping
- Encode categorical variables
- Feature scaling

Evaluation Metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Expected Goal:

- Achieve $\mathbf{R^2 \geq 0.80}$, suitable for mobile app deployment

Task 2: Unsupervised Learning – Workout Pattern Clustering

Objective:

Identify hidden workout behavior patterns **without using labels(`Workout_Type`)**, especially focusing on workout intensity and physiological response.

Steps:

1. Drop target variable (`Workout_Type`)
2. Encode categorical variables
3. Scale numerical features
4. Apply **PCA** for dimensionality reduction
5. Apply **KMeans Clustering**
6. Evaluate clustering quality using **Silhouette Score**

Additional Unsupervised Learning Methods (Optional for Comparison)

- Hierarchical Clustering
- DBSCAN (Density-Based Spatial Clustering)

Algorithms:

- PCA (for feature compression)
- KMeans Clustering

Evaluation Metric:

- **Silhouette Score**

Acceptance Criterion:

- **Silhouette Score ≥ 0.15** (considered valid due to real-world physiological overlap)

Interpretation:

- Analyze cluster centroids
- Explain workout intensity patterns
- Compare clusters across experience levels

Why Silhouette ≥ 0.15 Is Acceptable (Industry Context)

- Fitness data is **highly overlapping** (similar heart rates across workouts)
- Human physiology does not form sharply separable clusters
- Real-world behavioral data rarely exceeds silhouette scores of 0.3
- PCA compresses variance, improving interpretability over separation
- Clusters are **behaviorally meaningful**, not mathematically perfect

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Expected Outcomes

- Accurate calorie burn prediction model
- Identification of workout intensity clusters
- Clear understanding of feature importance
- Practical exposure to supervised + unsupervised ML
- Industry-aligned evaluation mindset

Project Evaluation Metrics

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Regression:

- MAE
- RMSE
- R^2 Score

Clustering:

- Silhouette Score
- Cluster size distribution
- Cluster feature means

Technical Tags

- Python
- Pandas
- NumPy
- Scikit-learn

- Matplotlib / Seaborn
- Regression Modeling
- Clustering
- PCA

Project Deliverables

1. **Source Code**
    - Well-documented Python notebooks or scripts
2. **Project Report**
    - Problem statement
    - EDA findings
    - Model comparison
    - Business insights
3. **Visualizations**
    - Feature distributions
    - Model performance plots
    - PCA cluster plots

Project Guidelines

Coding Standards:

- Clear variable naming
- Modular functions
- Inline comments

Version Control:

- Git-based repository
- Clean commit history

Validation:

- Train-test split
- Cross-validation
- Fixed random seeds

References:

Project Live Evaluation Guidelines	Project Live Evaluation
EDA Guide	Exploratory Data Analysis (EDA) G…
Capstone Explanation Guideline	Capstone Explanation Guideline
GitHub Reference	How to Use GitHub.pptx
Streamlit Reference English	Streamlit Recordings
Streamlit Reference Tamil	Streamlit Recording Tamil
Project Orientation (English)	
Project Orientation (Tamil)	Project Orientation Session : Fitbit : .

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.
Note: Book the slot at least before 12:00 Pm on the same day

Timing: Monday-Saturday (4:00PM to 5:00PM)

Booking link : PROJECT DOUBT CLARIFICATION SESSION

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing: Monday-Saturday (05:30PM to 07:00PM)

Booking link : https://forms.gle/1m2Gsro41fLtZurRA

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